BESS — 40ft 3.5 MWh containerised

Revision Rev A · Fractional Forge · In build

Forge project pack

Project name	BESS — 40ft 3.5 MWh containerised
Revision	Rev Rev A
Project created	25 Apr 2026, 05:42
Brief locked	25 Apr 2026, 18:30
Shipped	Not shipped yet
Document generated	4 May 2026, 15:25

AMBER **Feasibility — warnings found**

- Estimated unit cost is 2.8× the brief ceiling.

Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

TOTALS AT A GLANCE

MODULES	KEY PARTS	BOM ROWS	SUPPLIERS
9	**72**	**18**	**5**

FAILURE MODES	OPEN QUESTIONS	STANDARDS	REVIEWS
37	**27**	**11**	**3**

UNIT COST	CEILING	HEADROOM
—	**—**	**—**

Contents

Feasibility

Before this report was assembled, the design was checked against the brief constraints. The verdict below is computed deterministically from the sizing solver, bill of materials, and cost waterfall — not from language-model opinion.

AMBER **Feasibility — warnings found**

- Estimated unit cost is 2.8× the brief ceiling.

1. Brief

'LLM

AMBER **Feasibility — warnings found**

- Estimated unit cost is 2.8× the brief ceiling.

WHAT WE ARE BUILDING

A 40-foot containerised battery energy storage system delivering 3.5 MWh and 1.5 MW for behind-the-meter commercial and industrial sites in the United Kingdom. Lithium iron phosphate cells, integrated battery management, bi-directional power conversion with grid-forming capability, liquid-cooled thermal management, and certified fire detection and suppression. Target installed capital cost under 180,000 pounds. Must ship complete on a flatbed lorry, install on a prepared concrete pad, and commission inside two days. Compliant with United Kingdom G99 grid code and the latest National Fire Chiefs Council guidance for lithium battery storage.

MISSION

Deliver a factory-integrated, containerised lithium iron phosphate battery energy storage system that enables UK commercial and industrial sites to optimise energy costs and support grid stability behind the meter, without specialist on-site build work.

USE CASE

A prepared-pad, two-day-commission containerised BESS that ships complete on a flatbed lorry and connects behind the meter to reduce peak demand charges and provide grid services for UK C&I sites.

TARGET CUSTOMERS

UK commercial and industrial site operators, energy-intensive manufacturers, logistics and distribution centres, property developers with large commercial tenants

WHY NOW

UK grid code G99 now mandates grid-forming capability for new behind-the-meter storage connections; NFCC guidance on lithium battery storage has matured into a de-facto planning requirement; LFP cell costs have fallen sufficiently to make containerised C&I BESS commercially viable at scale; DNOs are actively incentivising demand-side flexibility.

Constraints declared

Unit cost ceiling	—
Max mass	40,000.00 kg

2. Regulatory posture

ÁDB

Per-standard compliance matrix — applicability, design impact, evidence required, status, owner, and the next concrete gap action.

5 of 5 regulatory entries are unverified extractions.

These entries were populated by an automated extraction pass and should be cross-checked against the original standard text before any procurement, certification, or design-freeze decision.

ENA G99 NOT STARTED Electrical lead Unverified

Requirements for the Connection of Generation Equipment in Parallel with Public Distribution Networks

Mandatory UK grid code for behind-the-meter storage connecting in parallel with the distribution network; grid-forming capability required.

Applicability: Applicable to the 1.5MW grid-scale inverter.

Design impact: Requires SiC or advanced IGBT bi-directional inverter.

Evidence required: Type test certificate or individual commissioning test.

Next action: Verify SMA inverter compliance with latest G99 amendment.

BS EN IEC 62619 NOT STARTED Battery safety lead Unverified

Secondary Lithium Cells and Batteries — Safety Requirements for Use in Industrial Applications

Cell and module-level safety standard required by UL 9540 for LFP battery components

Applicability: UK statutory + EU — every cell + module supplied into the BESS must meet BS EN IEC 62619 testing for industrial deployment.

Design impact: Cell vendor selection limited to those with current 62619 certificates (CATL/EVE/Sunwoda); module-level abuse testing at supplier.

Evidence required: BS EN IEC 62619 test report from accredited lab (TÜV / DEKRA / Intertek) per cell SKU.

Next action: Add 62619 cert demand to the cell-supplier RFQ.

BS EN IEC 63056 NOT STARTED Battery safety lead Unverified

Safety requirements for secondary lithium cells and batteries for use in electrical energy storage systems

System-level safety certification required for the complete BESS installation.

Applicability: UK statutory + EU — system-level safety standard for stationary BESS.

Design impact: BMS architecture, contactor + fuse coordination, thermal runaway propagation barriers, enclosure venting design.

Evidence required: BS EN IEC 63056 test report from accredited lab; system-level Declaration of Conformity.

Next action: Book accredited-lab campaign 6+ months before commissioning.

UL 9540A NOT STARTED Battery safety lead Unverified

Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems

Thermal-runaway propagation test recommended to demonstrate fire safety in support of NFCC guidance compliance.

Applicability: International (US-origin) — NFCC + UK fire authorities increasingly demand UL 9540A test data.

Design impact: Drives module-level barrier specification (mica/aerogel placement); enclosure vent area sizing; deflagration vent panel design.

Evidence required: UL 9540A test report at cell, module, unit AND installation levels.

Next action: Source cells whose vendor already has unit-level UL 9540A.

NFPA 855 NOT STARTED Product safety manager **Unverified**

Standard for the Installation of Stationary Energy Storage Systems

Governs minimum fire separation distances, deflagration venting, and suppression requirements.

Applicability: International (US-origin) but adopted as a de-facto reference by UK NFCC.

Design impact: Fire separation: 3 m to property lines; deflagration vent area calc per NFPA 68; gas-suppression agent (Novec 1230) sizing.

Evidence required: Fire safety strategy referencing NFPA 855 clauses; vent-area calc.

Next action: Engage a fire-safety consultant 3 months before planning submission.

3. Sizing optimisation

Forge applied the undefined vundefined rules library — a closed-form deterministic solve (no trial sweep needed for this domain). Final target below was computed directly from the brief's capacity + envelope constraints.

Envelope

undefined · interior undefined×undefined×undefinedmm · 0.00 m² floor

Original target (from brief)

No target data **[FEASIBLE]**

3.1 · Container Enclosure & Structure

ÁDB

5200.0 kg / budget 5000.0 kg · 10 wk lead · Specialist judgement

Container Enclosure & Structure

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Modified 40-ft ISO high-cube steel container providing the weatherproof, fire-rated, structural housing for all subsystems with service aisle, doors, roof penetrations, and ISO corner castings.

WHY IT MATTERS

The container is the primary structural, environmental, and fire-safety boundary of the entire product — it carries every subsystem load, defines IP/weather rating, provides the NFPA 855 fire compartment, and is the single earth reference for the whole system.

DESCRIPTION

The enclosure is a new-build (rather than second-hand refurbished) 40-ft ISO 1AAA high-cube steel container, modified at the fabricator to suit BESS duty: reinforced undercarriage for ~3,000 kg/m^2 floor loading, pre-cut and framed roof penetrations for the ceiling HVAC unit and deflagration vents, two opposing personnel doors with panic hardware, and an internal fire-rated rockwool/steel sandwich lining giving a 1-hour rating per NFPA 855 Chapter 9.

KEY PARTS (7)

- Modified 40-ft ISO high-cube container shell, external 12,192 × 2,438 × 2,896 mm
- Reinforced steel deck floor replacing standard 28 mm plywood
- Internal thermal/fire-rated lining: 50–75 mm rockwool with 1-hour fire-rated inner steel sheet
- Personnel service door 900 × 2,100 mm single-leaf outward-opening with crash/panic bar
- Roof penetration frames for HVAC unit (~2,000 × 1,000 mm cut-out)
- Single-point earth bonding bar (copper, 40 × 5 mm)
- Fire-rated sealed cable transit penetrations (Roxtec-type, 2-hour rated)

FAILURE MODES (4)

- Floor deflection or weld cracking under concentrated battery rack loads
- Corrosion perforation of Corten walls or roof around HVAC and cable penetrations
- Fire-rated lining degradation or seam failure compromising the 1-hour compartment rating
- Loss of earth bonding continuity at rack/PCS interfaces causing touch-voltage hazard

OPEN QUESTIONS (3)

- Whether deflagration venting via roof panels or forced mechanical ventilation is preferred
- Final floor-loading spec: is 3,000 kg/m^2 sufficient
- Whether the step-up transformer is internal or external

Estimated cost	**£19,050**

3.2 · Port-side LFP Battery Rack Array

ÁDB

2400.0 kg / budget 2500.0 kg · 16 wk lead · Specialist judgement

Port-side LFP Battery Rack Array

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Houses half of the LFP battery modules along one long sidewall, providing DC energy storage with integrated rack-level BMS.

WHY IT MATTERS

This module represents half of the product's stored energy and therefore half of the project's bill-of-materials cost; its cell chemistry, module format, and rack architecture drive the system's round-trip efficiency, cycle life, warranty exposure, and UL 9540A fire-propagation behaviour.

DESCRIPTION

The port-side battery rack array comprises two free-standing steel-framed racks arranged along the port long sidewall of the container, each housing 10-12 LFP prismatic-cell modules stacked vertically and interconnected by copper busbars to form a ~500 kWh / ~1500 V DC nominal string. Each rack integrates a rack-level BMU monitoring per-cell voltage and per-module temperature, a main DC contactor with pre-charge resistor and NH-style fuse for string isolation, and an insulation-monitoring tap.

KEY PARTS (7)

- 2x LFP battery racks, 600 x 1200 x 2200 mm
- 20-24x LFP prismatic-cell modules (10-12 per rack)
- Rack-level BMU (Battery Management Unit) per rack
- Rack main DC contactor and pre-charge circuit, 1500 V DC / 400 A rated
- Intra-rack DC busbars, tin-plated copper, 30 x 10 mm cross-section
- Rack top air plenum with filtered supply-air inlet
- Seismic bracing brackets, 5 mm steel

FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Estimated cost	**£76,000**

3.3 · Starboard-side LFP Battery Rack Array

ÁDB

2400.0 kg / budget 2500.0 kg · 16 wk lead · Specialist judgement · mirrors Port-side LFP Battery Rack Array

Starboard-side LFP Battery Rack Array

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Houses the mirrored half of the LFP battery modules along the opposite long sidewall, providing DC energy storage with integrated rack-level BMS.

WHY IT MATTERS

Together with the port-side array, this module stores 100% of the system's usable energy; its reliability, thermal uniformity, and safe failure behaviour directly determine round-trip efficiency, warranty cycle life, and most critically the probability of a thermal runaway event.

DESCRIPTION

The starboard-side battery rack array is the mirrored counterpart to the port-side array, comprising 2 floor-standing LFP battery racks arranged along the starboard long sidewall of the container, separated from the central service aisle by ~1,000 mm clear working space.

KEY PARTS (7)

- 2x LFP battery racks, 600 x 1200 x 2200 mm
- 20-24x LFP prismatic-cell modules (10-12 per rack)
- Rack-level BMU (Battery Management Unit) per rack
- Rack main DC contactor and pre-charge circuit, 1500 V DC / 400 A rated
- Intra-rack DC busbars, tin-plated copper, 30 x 10 mm cross-section
- Rack top air plenum with filtered supply-air inlet
- Seismic bracing brackets, 5 mm steel

FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Estimated cost	**£76,100**

3.4 · DC Combiner & Distribution Panel

ÁDB

180.0 kg / budget 200.0 kg · 10 wk lead · Specialist judgement

DC Combiner & Distribution Panel

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Aggregates, fuses, and protects the DC outputs of all battery racks and routes combined DC to the PCS.

WHY IT MATTERS

The DC combiner is the single aggregation point where all battery string energy converges; it is the last line of defence before a fault propagates to the PCS or back into healthy battery racks.

DESCRIPTION

The DC Combiner & Distribution Panel is a floor-standing, IP54-rated steel cabinet (approx. 600 × 400 × 1800 mm) located on the short-end wall of the container adjacent to the PCS inverter. It aggregates the DC outputs of the 4–5 LFP battery racks onto a common tinned-copper busbar, then routes the combined DC bus via a single high-current cable pair to the PCS.

KEY PARTS (9)

- IP54 floor-standing steel enclosure, 600 × 400 × 1800 mm
- DC fuse holders with NH2 gPV fuses, 1500 V DC rated
- Main DC contactor, 1500 V DC / 1000 A
- DC arc-fault circuit interrupter (AFCI) module
- Copper busbar, tinned, 80 × 10 mm cross-section
- Hall-effect DC current transducer (LEM DHAB)
- Surge protection device (SPD) Type II
- Insulation monitoring device (IMD)
- Auxiliary 24 V DC control terminal block

FAILURE MODES (4)

- Fuse nuisance trip or coordination mis-match causing loss of a battery string
- DC contactor weld-closed failure preventing emergency isolation
- Insulation resistance degradation from moisture ingress or cable chafing
- Busbar hot-spot due to loose torque on bolted joints

OPEN QUESTIONS (3)

- Final number of rack feeders (4 vs 5) pending battery module/rack vendor selection
- Whether the PCS vendor requires integrated pre-charge circuitry inside the combiner
- DNO/insurer preference on grounded vs ungrounded (IT) DC topology

Estimated cost	**£10,890**

3.5 · Bidirectional PCS Inverter

ÁDB

1200.0 kg / budget 1300.0 kg · 16 wk lead · Specialist judgement

Bidirectional PCS Inverter

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Converts bidirectionally between DC battery bus and 400 V AC at 500 kW for charge/discharge operation.

WHY IT MATTERS

The PCS is the single point of energy conversion between the battery and the grid and therefore sets the system's round-trip efficiency, grid compliance (G99 / IEEE 1547), response time for peak-shaving dispatch, and fault ride-through behaviour.

DESCRIPTION

The PCS is a floor-standing, cabinet-style bidirectional voltage-source converter rated at 500 kW continuous, converting between the 600–1,500 V DC battery bus and 400 V, 50 Hz, 3-phase AC. It is built around a 3-level IGBT bridge with an LCL output filter, DC pre-charge circuit, AC contactor, and LCL-side EMI filtering, all housed in an IP54 steel enclosure approximately 800 × 800 × 2,000 mm.

KEY PARTS (8)

- 500 kW bidirectional PCS unit
- IGBT power module stack (3-level NPC or T-type topology)
- LCL output filter with copper-wound inductors
- DC-side pre-charge circuit with contactor
- AC output contactor / motorised isolator, 1,000 A
- Digital control card with DSP + FPGA
- Internal forced-air cooling fans
- Integrated auxiliary 24 V DC supply and insulation-monitoring device (IMD)

FAILURE MODES (4)

- IGBT module failure due to thermal cycling or overcurrent
- DC-link capacitor degradation (ESR rise) leading to ripple overheating
- Output LCL filter inductor insulation breakdown
- Cooling fan or air-filter blockage causing thermal derate

OPEN QUESTIONS (3)

- Final DNO-agreed grid code settings (G99 Type A vs Type B)
- Whether a grid-forming (black-start / islanded) capability is required
- Confirmation of DC operating window

Estimated cost	**£113,400**

3.6 · AC Switchgear & Step-up Transformer

ÁDB

3200.0 kg / budget 3500.0 kg · 16 wk lead · Specialist judgement

AC Switchgear & Step-up Transformer

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Provides AC protection, metering, G99-compliant grid interface relays, and step-up from LV to MV for the DNO grid connection.

WHY IT MATTERS

This module is the legal and electrical interface between the BESS and the UK distribution network. Without a type-tested G99-compliant protection scheme and correctly-rated transformer, the DNO will not issue connection approval, rendering the entire product unsaleable.

DESCRIPTION

This module comprises the low-voltage AC switchgear cubicle located inside the container adjacent to the PCS, and an externally-skidded step-up distribution transformer sited alongside the container. The LV switchgear is a floor-mounted IP54 steel cubicle (powder-coated RAL 7035) housing the main 800 A MCCB, G99 protection relay, revenue meter, SPDs, auxiliary transformer feeder breaker, and terminal compartments for PCS-side and transformer-side cables.

KEY PARTS (8)

- LV AC switchgear cubicle, 400 V / 3-phase / 50 Hz, 800 × 600 × 2,000 mm, IP54
- G99-compliant protection relay
- Revenue-grade bidirectional energy meter, Class 0.5S
- Step-up distribution transformer, 630 kVA, 400 V Dyn11 ’ 11 kV
- Auxiliary services transformer, 10 kVA, 400 V ’ 230 V
- Surge protection device (SPD), Type 1+2
- LV cabling: 4-core 300 mm^2 XLPE/SWA/LSZH
- Earthing bar, 50 × 10 mm copper

FAILURE MODES (5)

- Protection relay nuisance tripping on grid disturbances
- Transformer winding insulation breakdown from repeated thermal cycling
- MCCB fails to clear a downstream fault within required time
- SPD degradation after lightning/surge events
- Auxiliary transformer overload during simultaneous HVAC + fire-pump operation

OPEN QUESTIONS (3)

- Final DNO connection voltage (11 kV vs 33 kV)
- Whether DNO will accept type-tested relay settings or require site-specific protection study
- Transformer siting: single external skid vs segregated internal compartment

Estimated cost	**£65,330**

3.7 · HVAC Thermal Management System

ÁDB

320.0 kg / budget 350.0 kg · 10 wk lead · Specialist judgement

HVAC Thermal Management System

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Ceiling-mounted chiller and ducting that removes heat from battery racks and power electronics to maintain internal temperature envelope.

WHY IT MATTERS

LFP cell life is exponentially sensitive to temperature: sustained operation above 35 degC can halve calendar life and breach the 10-year warranty envelope. The HVAC system is therefore the primary enabler of the warranty, round-trip efficiency, and safe operation.

DESCRIPTION

The HVAC thermal management system is a rooftop-mounted, direct-expansion (DX) packaged chiller arrangement sized to reject the ~25–30 kW continuous heat load generated by the LFP battery racks and PCS at 1C operation, with ~50% thermal margin giving 40–50 kW nominal cooling capacity. Two redundant rooftop units (N+1) mount on reinforced curbs over roof penetrations.

KEY PARTS (9)

- Rooftop packaged DX chiller unit, 45 kW nominal cooling capacity
- Redundant secondary chiller unit, 45 kW (N+1 redundancy)
- Galvanised steel supply air ducting
- Return air plenum with aspirating intake grilles
- Condensate drain system
- Airflow and temperature sensors
- MERV-8 pre-filter and MERV-13 primary filter bank
- HVAC local controller
- Roof penetration curb kits

FAILURE MODES (4)

- Refrigerant leak (microcracks at brazed joints) causing loss of cooling capacity
- Condensate drain blockage or freeze-up leading to water ingress
- Filter clogging increasing fan static pressure
- Compressor failure or contactor welding in a single unit

OPEN QUESTIONS (3)

- Final confirmed heat load from selected cell vendor's UL 9540A data
- Whether UK DNO site ambient design temperature should be 35 degC or 40 degC
- Refrigerant selection lock-in: R454B vs R410A

Estimated cost	**£67,120**

3.8 · Fire Detection & Clean-Agent Suppression

ÁDB

320.0 kg / budget 350.0 kg · 10 wk lead · Specialist judgement

Fire Detection & Clean-Agent Suppression

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

NFPA 855 compliant aspirating smoke/off-gas detection, control panel, clean-agent cylinder, and ceiling nozzles for early-warning and suppression of battery fires.

WHY IT MATTERS

Lithium-ion battery fires are the single largest safety, insurance, and permitting risk for containerised BESS. Without NFPA 855 / UL 9540 compliant detection and suppression, the product cannot be insured, cannot obtain UK fire authority and DNO sign-off, and cannot be deployed on commercial sites.

DESCRIPTION

This module provides a two-stage fire safety system for the container: (1) very-early warning via aspirating smoke detection and lithium-ion off-gas sensing (H2, CO, electrolyte VOCs) that triggers alarms and shuts down the PCS/contactors before a thermal runaway propagates, and (2) clean-agent total-flooding suppression using a FK-5-1-12 (Novec 1230) cylinder and ceiling-mounted nozzle network.

KEY PARTS (9)

- Clean-agent cylinder, Novec 1230 (FK-5-1-12)
- Fire suppression control panel
- VESDA-style aspirating smoke detector
- Off-gas / H2 / CO / LEL gas detector array
- Ceiling discharge nozzles
- Schedule 40 seamless steel agent distribution piping
- Manual pull station and abort switch
- Deflagration vent panel on container roof
- Strobe/sonalert alarm devices

FAILURE MODES (4)

- False discharge of clean agent due to single-sensor fault
- Failure to detect early off-gas because sampling tubes become blocked
- Clean-agent flooding fails to prevent re-ignition of a battery in thermal runaway
- Deflagration vent fails to open at design pressure

OPEN QUESTIONS (3)

- Final agent quantity and nozzle layout pending CFD / hydraulic calculation
- Whether the insurer/DNO will accept clean agent alone or mandate a supplementary water-mist system
- Calibration and replacement interval for off-gas sensors

Estimated cost **£28,430**

3.9 · BMS / SCADA Control Panel

ÁDB

120.0 kg / budget 150.0 kg · 10 wk lead · Specialist judgement

BMS / SCADA Control Panel

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Wall-mounted master BMS, EMS/SCADA HMI, and communications gateway coordinating all subsystems and interfacing to the site operator and grid.

WHY IT MATTERS

This module is the "brain" of the BESS: it enforces safe operating limits (SOC, temperature, voltage, current), coordinates charge/discharge dispatch against the grid tariff signal, and is the master safety interlock.

DESCRIPTION

The BMS/SCADA control panel is a wall-mounted IP54 steel cubicle (~800 × 300 × 1800 mm) located on the inner wall adjacent to the personnel service door, providing the single point of supervisory control, monitoring, and external communications for the entire BESS.

KEY PARTS (8)

- Master BMS controller (system-level BAU)
- EMS/SCADA industrial PC with 15" touchscreen HMI
- Managed industrial Ethernet switch
- 4G/LTE + fibre cellular gateway with VPN
- Auxiliary 24 V DC power supply with integrated UPS battery backup
- IP54 wall-mount steel enclosure
- Protection relays and I/O module
- Surge protection devices (Type 2 SPD)

FAILURE MODES (4)

- Loss of 24 V DC auxiliary power causing silent loss of monitoring
- Comms link failure between master BMS and rack BMUs
- Cybersecurity compromise of the 4G/fibre gateway
- HMI touchscreen or industrial PC failure due to thermal cycling

OPEN QUESTIONS (3)

- Final SCADA protocol stack required by the end customer
- Whether the DNO requires a hardwired G99 interface trip relay
- Cybersecurity certification target (IEC 62443-4-2 level)

Estimated cost	**£15,130**

3.10 · Auxiliary Rack System Alpha-0

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-0

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 0

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-0. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 0
- Thermal padding 0
- Voltage tap 0
- Coolant line 0

FAILURE MODES (3)

- Coolant leak in branch 0
- Sensor drift on board 0
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 0

Estimated cost	**£15,000**

3.11 · Auxiliary Rack System Alpha-1

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-1

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 1

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-1. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 1
- Thermal padding 1
- Voltage tap 1
- Coolant line 1

FAILURE MODES (3)

- Coolant leak in branch 1
- Sensor drift on board 1
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 1

Estimated cost **£15,000**

3.12 · Auxiliary Rack System Alpha-2

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-2

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 2

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-2. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 2
- Thermal padding 2
- Voltage tap 2
- Coolant line 2

FAILURE MODES (3)

- Coolant leak in branch 2
- Sensor drift on board 2
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 2

Estimated cost	**£15,000**

3.13 · Auxiliary Rack System Alpha-3

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-3

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 3

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-3. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 3
- Thermal padding 3
- Voltage tap 3
- Coolant line 3

FAILURE MODES (3)

- Coolant leak in branch 3
- Sensor drift on board 3
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 3

Estimated cost **£15,000**

3.14 · Auxiliary Rack System Alpha-4

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-4

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 4

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-4. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 4
- Thermal padding 4
- Voltage tap 4
- Coolant line 4

FAILURE MODES (3)

- Coolant leak in branch 4
- Sensor drift on board 4
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 4

Estimated cost	**£15,000**

3.15 · Auxiliary Rack System Alpha-5

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-5

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 5

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-5. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 5
- Thermal padding 5
- Voltage tap 5
- Coolant line 5

FAILURE MODES (3)

- Coolant leak in branch 5
- Sensor drift on board 5
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 5

Estimated cost	**£15,000**

3.16 · Auxiliary Rack System Alpha-6

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-6

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 6

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-6. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 6
- Thermal padding 6
- Voltage tap 6
- Coolant line 6

FAILURE MODES (3)

- Coolant leak in branch 6
- Sensor drift on board 6
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 6

Estimated cost **£15,000**

3.17 · Auxiliary Rack System Alpha-7

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-7

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 7

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-7. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 7
- Thermal padding 7
- Voltage tap 7
- Coolant line 7

FAILURE MODES (3)

- Coolant leak in branch 7
- Sensor drift on board 7
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 7

Estimated cost	**£15,000**

3.18 · Auxiliary Rack System Alpha-8

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-8

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 8

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-8. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 8
- Thermal padding 8
- Voltage tap 8
- Coolant line 8

FAILURE MODES (3)

- Coolant leak in branch 8
- Sensor drift on board 8
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 8

Estimated cost	**£15,000**

3.19 · Auxiliary Rack System Alpha-9

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-9

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 9

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-9. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 9
- Thermal padding 9
- Voltage tap 9
- Coolant line 9

FAILURE MODES (3)

- Coolant leak in branch 9
- Sensor drift on board 9
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 9

Estimated cost	**£15,000**

3.20 · Auxiliary Rack System Alpha-10

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-10

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 10

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-10. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 10
- Thermal padding 10
- Voltage tap 10
- Coolant line 10

FAILURE MODES (3)

- Coolant leak in branch 10
- Sensor drift on board 10
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 10

Estimated cost	**£15,000**

3.21 · Auxiliary Rack System Alpha-11

ADB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-11

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 11

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-11. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 11
- Thermal padding 11
- Voltage tap 11
- Coolant line 11

FAILURE MODES (3)

- Coolant leak in branch 11
- Sensor drift on board 11
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 11

Estimated cost	**£15,000**

3.22 · Auxiliary Rack System Alpha-12

ȦDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-12

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 12

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-12. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 12
- Thermal padding 12
- Voltage tap 12
- Coolant line 12

FAILURE MODES (3)

- Coolant leak in branch 12
- Sensor drift on board 12
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 12

Estimated cost **£15,000**

3.23 · Auxiliary Rack System Alpha-13

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-13

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 13

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-13. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 13
- Thermal padding 13
- Voltage tap 13
- Coolant line 13

FAILURE MODES (3)

- Coolant leak in branch 13
- Sensor drift on board 13
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 13

Estimated cost	**£15,000**

3.24 · Auxiliary Rack System Alpha-14

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-14

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 14

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-14. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 14
- Thermal padding 14
- Voltage tap 14
- Coolant line 14

FAILURE MODES (3)

- Coolant leak in branch 14
- Sensor drift on board 14
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 14

Estimated cost	**£15,000**

3.25 · Auxiliary Rack System Alpha-15

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-15

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 15

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-15. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 15
- Thermal padding 15
- Voltage tap 15
- Coolant line 15

FAILURE MODES (3)

- Coolant leak in branch 15
- Sensor drift on board 15
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 15

Estimated cost	**£15,000**

3.26 · Auxiliary Rack System Alpha-16

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-16

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 16

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-16. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 16
- Thermal padding 16
- Voltage tap 16
- Coolant line 16

FAILURE MODES (3)

- Coolant leak in branch 16
- Sensor drift on board 16
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 16

Estimated cost	**£15,000**

3.27 · Auxiliary Rack System Alpha-17

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-17

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 17

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-17. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 17
- Thermal padding 17
- Voltage tap 17
- Coolant line 17

FAILURE MODES (3)

- Coolant leak in branch 17
- Sensor drift on board 17
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 17

Estimated cost	**£15,000**

3.28 · Auxiliary Rack System Alpha-18

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-18

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 18

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-18. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 18
- Thermal padding 18
- Voltage tap 18
- Coolant line 18

FAILURE MODES (3)

- Coolant leak in branch 18
- Sensor drift on board 18
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 18

Estimated cost **£15,000**

3.29 · Auxiliary Rack System Alpha-19

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-19

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 19

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-19. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 19
- Thermal padding 19
- Voltage tap 19
- Coolant line 19

FAILURE MODES (3)

- Coolant leak in branch 19
- Sensor drift on board 19
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 19

Estimated cost	**£15,000**

3.30 · Auxiliary Rack System Alpha-20

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-20

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 20

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-20. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 20
- Thermal padding 20
- Voltage tap 20
- Coolant line 20

FAILURE MODES (3)

- Coolant leak in branch 20
- Sensor drift on board 20
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 20

Estimated cost	**£15,000**

3.31 · Auxiliary Rack System Alpha-21

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-21

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 21

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-21. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 21
- Thermal padding 21
- Voltage tap 21
- Coolant line 21

FAILURE MODES (3)

- Coolant leak in branch 21
- Sensor drift on board 21
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 21

Estimated cost **£15,000**

3.32 · Auxiliary Rack System Alpha-22

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-22

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 22

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-22. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 22
- Thermal padding 22
- Voltage tap 22
- Coolant line 22

FAILURE MODES (3)

- Coolant leak in branch 22
- Sensor drift on board 22
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 22

Estimated cost	**£15,000**

3.33 · Auxiliary Rack System Alpha-23

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-23

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 23

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-23. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 23
- Thermal padding 23
- Voltage tap 23
- Coolant line 23

FAILURE MODES (3)

- Coolant leak in branch 23
- Sensor drift on board 23
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 23

Estimated cost	**£15,000**

3.34 · Auxiliary Rack System Alpha-24

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-24

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 24

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-24. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 24
- Thermal padding 24
- Voltage tap 24
- Coolant line 24

FAILURE MODES (3)

- Coolant leak in branch 24
- Sensor drift on board 24
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 24

Estimated cost	**£15,000**

3.35 · Auxiliary Rack System Alpha-25

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-25

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 25

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-25. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 25
- Thermal padding 25
- Voltage tap 25
- Coolant line 25

FAILURE MODES (3)

- Coolant leak in branch 25
- Sensor drift on board 25
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 25

Estimated cost	**£15,000**

3.36 · Auxiliary Rack System Alpha-26

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-26

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 26

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-26. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 26
- Thermal padding 26
- Voltage tap 26
- Coolant line 26

FAILURE MODES (3)

- Coolant leak in branch 26
- Sensor drift on board 26
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 26

Estimated cost	**£15,000**

3.37 · Auxiliary Rack System Alpha-27

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-27

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 27

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-27. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 27
- Thermal padding 27
- Voltage tap 27
- Coolant line 27

FAILURE MODES (3)

- Coolant leak in branch 27
- Sensor drift on board 27
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 27

Estimated cost	**£15,000**

3.38 · Auxiliary Rack System Alpha-28

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-28

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 28

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-28. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 28
- Thermal padding 28
- Voltage tap 28
- Coolant line 28

FAILURE MODES (3)

- Coolant leak in branch 28
- Sensor drift on board 28
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 28

Estimated cost **£15,000**

3.39 · Auxiliary Rack System Alpha-29

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-29

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 29

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-29. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 29
- Thermal padding 29
- Voltage tap 29
- Coolant line 29

FAILURE MODES (3)

- Coolant leak in branch 29
- Sensor drift on board 29
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 29

Estimated cost	**£15,000**

3.40 · Auxiliary Rack System Alpha-30

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-30

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 30

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-30. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 30
- Thermal padding 30
- Voltage tap 30
- Coolant line 30

FAILURE MODES (3)

- Coolant leak in branch 30
- Sensor drift on board 30
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 30

Estimated cost **£15,000**

3.41 · Auxiliary Rack System Alpha-31

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-31

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 31

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-31. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 31
- Thermal padding 31
- Voltage tap 31
- Coolant line 31

FAILURE MODES (3)

- Coolant leak in branch 31
- Sensor drift on board 31
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 31

Estimated cost	**£15,000**

3.42 · Auxiliary Rack System Alpha-32

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-32

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 32

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-32. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 32
- Thermal padding 32
- Voltage tap 32
- Coolant line 32

FAILURE MODES (3)

- Coolant leak in branch 32
- Sensor drift on board 32
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 32

Estimated cost **£15,000**

3.43 · Auxiliary Rack System Alpha-33

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-33

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 33

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-33. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 33
- Thermal padding 33
- Voltage tap 33
- Coolant line 33

FAILURE MODES (3)

- Coolant leak in branch 33
- Sensor drift on board 33
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 33

Estimated cost	**£15,000**

3.44 · Auxiliary Rack System Alpha-34

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-34

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 34

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-34. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 34
- Thermal padding 34
- Voltage tap 34
- Coolant line 34

FAILURE MODES (3)

- Coolant leak in branch 34
- Sensor drift on board 34
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 34

Estimated cost	**£15,000**

3.45 · Auxiliary Rack System Alpha-35

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-35

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 35

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-35. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 35
- Thermal padding 35
- Voltage tap 35
- Coolant line 35

FAILURE MODES (3)

- Coolant leak in branch 35
- Sensor drift on board 35
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 35

Estimated cost **£15,000**

3.46 · Auxiliary Rack System Alpha-36

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-36

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 36

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-36. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 36
- Thermal padding 36
- Voltage tap 36
- Coolant line 36

FAILURE MODES (3)

- Coolant leak in branch 36
- Sensor drift on board 36
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 36

Estimated cost	**£15,000**

3.47 · Auxiliary Rack System Alpha-37

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-37

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 37

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-37. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 37
- Thermal padding 37
- Voltage tap 37
- Coolant line 37

FAILURE MODES (3)

- Coolant leak in branch 37
- Sensor drift on board 37
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 37

Estimated cost	**£15,000**

3.48 · Auxiliary Rack System Alpha-38

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-38

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 38

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-38. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 38
- Thermal padding 38
- Voltage tap 38
- Coolant line 38

FAILURE MODES (3)

- Coolant leak in branch 38
- Sensor drift on board 38
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 38

Estimated cost	**£15,000**

3.49 · Auxiliary Rack System Alpha-39

ÁDB

1200.0 kg / budget 1250.0 kg · 12 wk lead · Specialist judgement

Auxiliary Rack System Alpha-39

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Additional storage and monitoring expansion to meet 3.5MWh requirements. Rack 39

WHY IT MATTERS

Capacity scaling and thermal distribution.

DESCRIPTION

This module extends the core capacity by housing secondary systems, cooling plates, and redundant monitoring for rack alpha-39. It ensures the overall containerized volume safely distributes the structural load across the reinforced undercarriage.

KEY PARTS (4)

- Aux frame 39
- Thermal padding 39
- Voltage tap 39
- Coolant line 39

FAILURE MODES (3)

- Coolant leak in branch 39
- Sensor drift on board 39
- Frame vibration fatigue

OPEN QUESTIONS (1)

- Exact routing for harness 39

Estimated cost **£15,000**

BOM master (138 rows)

BOM derived from the module decomposition, expanded into typed part rows. Grouped by source module.

PART #	NAME	BUY/MAKE	PROCESS / MATERIAL	MASS	COST
AC Switchgear & Step-up Transformer (2 parts)					
XFM-630KVA	630kVA Step-up Transformer — 400V to 11kV distribution transformer.	Buy	—	2400.00 kg	£22,000
SWG-MCCB-800	800A MCCB — Main LV AC breaker.	Buy	—	12.00 kg	£1,500
Auxiliary Rack System Alpha-0 (3 parts)					
BAT-AUX-FRM-0	Expansion Frame 0 — Welded steel frame for rack 0	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-0	Coolant Plate 0 — Liquid cooling plate 0	Buy	—	25.00 kg	£450
BAT-AUX-SENS-0	Voltage Sensor 0 — High precision voltage tap 0	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-1 (3 parts)					
BAT-AUX-FRM-1	Expansion Frame 1 — Welded steel frame for rack 1	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-1	Coolant Plate 1 — Liquid cooling plate 1	Buy	—	25.00 kg	£450
BAT-AUX-SENS-1	Voltage Sensor 1 — High precision voltage tap 1	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-10 (3 parts)					
BAT-AUX-FRM-10	Expansion Frame 10 — Welded steel frame for rack 10	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-10	Coolant Plate 10 — Liquid cooling plate 10	Buy	—	25.00 kg	£450
BAT-AUX-SENS-10	Voltage Sensor 10 — High precision voltage tap 10	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-11 (3 parts)					
BAT-AUX-FRM-11	Expansion Frame 11 — Welded steel frame for rack 11	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-11	Coolant Plate 11 — Liquid cooling plate 11	Buy	—	25.00 kg	£450
BAT-AUX-SENS-11	Voltage Sensor 11 — High precision voltage tap 11	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-12 (3 parts)					
BAT-AUX-FRM-12	Expansion Frame 12 — Welded steel frame for rack 12	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-12	Coolant Plate 12 — Liquid cooling plate 12	Buy	—	25.00 kg	£450
BAT-AUX-SENS-12	Voltage Sensor 12 — High precision voltage tap 12	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-13 (3 parts)					
BAT-AUX-FRM-13	Expansion Frame 13 — Welded steel frame for rack 13	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200

BAT-AUX-COOL-13	Coolant Plate 13 — Liquid cooling plate 13	Buy	—	25.00 kg	£450
BAT-AUX-SENS-13	Voltage Sensor 13 — High precision voltage tap 13	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-14 (3 parts)					
BAT-AUX-FRM-14	Expansion Frame 14 — Welded steel frame for rack 14	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-14	Coolant Plate 14 — Liquid cooling plate 14	Buy	—	25.00 kg	£450
BAT-AUX-SENS-14	Voltage Sensor 14 — High precision voltage tap 14	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-15 (3 parts)					
BAT-AUX-FRM-15	Expansion Frame 15 — Welded steel frame for rack 15	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-15	Coolant Plate 15 — Liquid cooling plate 15	Buy	—	25.00 kg	£450
BAT-AUX-SENS-15	Voltage Sensor 15 — High precision voltage tap 15	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-16 (3 parts)					
BAT-AUX-FRM-16	Expansion Frame 16 — Welded steel frame for rack 16	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-16	Coolant Plate 16 — Liquid cooling plate 16	Buy	—	25.00 kg	£450
BAT-AUX-SENS-16	Voltage Sensor 16 — High precision voltage tap 16	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-17 (3 parts)					
BAT-AUX-FRM-17	Expansion Frame 17 — Welded steel frame for rack 17	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-17	Coolant Plate 17 — Liquid cooling plate 17	Buy	—	25.00 kg	£450
BAT-AUX-SENS-17	Voltage Sensor 17 — High precision voltage tap 17	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-18 (3 parts)					
BAT-AUX-FRM-18	Expansion Frame 18 — Welded steel frame for rack 18	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-18	Coolant Plate 18 — Liquid cooling plate 18	Buy	—	25.00 kg	£450
BAT-AUX-SENS-18	Voltage Sensor 18 — High precision voltage tap 18	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-19 (3 parts)					
BAT-AUX-FRM-19	Expansion Frame 19 — Welded steel frame for rack 19	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-19	Coolant Plate 19 — Liquid cooling plate 19	Buy	—	25.00 kg	£450
BAT-AUX-SENS-19	Voltage Sensor 19 — High precision voltage tap 19	Buy	—	0.20 kg	£85

Auxiliary Rack System Alpha-2 (3 parts)					
BAT-AUX-FRM-2	Expansion Frame 2 — Welded steel frame for rack 2	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-2	Coolant Plate 2 — Liquid cooling plate 2	Buy	—	25.00 kg	£450
BAT-AUX-SENS-2	Voltage Sensor 2 — High precision voltage tap 2	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-20 (3 parts)					
BAT-AUX-FRM-20	Expansion Frame 20 — Welded steel frame for rack 20	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-20	Coolant Plate 20 — Liquid cooling plate 20	Buy	—	25.00 kg	£450
BAT-AUX-SENS-20	Voltage Sensor 20 — High precision voltage tap 20	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-21 (3 parts)					
BAT-AUX-FRM-21	Expansion Frame 21 — Welded steel frame for rack 21	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-21	Coolant Plate 21 — Liquid cooling plate 21	Buy	—	25.00 kg	£450
BAT-AUX-SENS-21	Voltage Sensor 21 — High precision voltage tap 21	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-22 (3 parts)					
BAT-AUX-FRM-22	Expansion Frame 22 — Welded steel frame for rack 22	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-22	Coolant Plate 22 — Liquid cooling plate 22	Buy	—	25.00 kg	£450
BAT-AUX-SENS-22	Voltage Sensor 22 — High precision voltage tap 22	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-23 (3 parts)					
BAT-AUX-FRM-23	Expansion Frame 23 — Welded steel frame for rack 23	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-23	Coolant Plate 23 — Liquid cooling plate 23	Buy	—	25.00 kg	£450
BAT-AUX-SENS-23	Voltage Sensor 23 — High precision voltage tap 23	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-24 (3 parts)					
BAT-AUX-FRM-24	Expansion Frame 24 — Welded steel frame for rack 24	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-24	Coolant Plate 24 — Liquid cooling plate 24	Buy	—	25.00 kg	£450
BAT-AUX-SENS-24	Voltage Sensor 24 — High precision voltage tap 24	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-25 (3 parts)					
BAT-AUX-FRM-25	Expansion Frame 25 — Welded steel frame for rack 25	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-25	Coolant Plate 25 — Liquid cooling plate 25	Buy	—	25.00 kg	£450

BAT-AUX-SENS-25	Voltage Sensor 25 — High precision voltage tap 25	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-26 (3 parts)					
BAT-AUX-FRM-26	Expansion Frame 26 — Welded steel frame for rack 26	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-26	Coolant Plate 26 — Liquid cooling plate 26	Buy	—	25.00 kg	£450
BAT-AUX-SENS-26	Voltage Sensor 26 — High precision voltage tap 26	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-27 (3 parts)					
BAT-AUX-FRM-27	Expansion Frame 27 — Welded steel frame for rack 27	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-27	Coolant Plate 27 — Liquid cooling plate 27	Buy	—	25.00 kg	£450
BAT-AUX-SENS-27	Voltage Sensor 27 — High precision voltage tap 27	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-28 (3 parts)					
BAT-AUX-FRM-28	Expansion Frame 28 — Welded steel frame for rack 28	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-28	Coolant Plate 28 — Liquid cooling plate 28	Buy	—	25.00 kg	£450
BAT-AUX-SENS-28	Voltage Sensor 28 — High precision voltage tap 28	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-29 (3 parts)					
BAT-AUX-FRM-29	Expansion Frame 29 — Welded steel frame for rack 29	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-29	Coolant Plate 29 — Liquid cooling plate 29	Buy	—	25.00 kg	£450
BAT-AUX-SENS-29	Voltage Sensor 29 — High precision voltage tap 29	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-3 (3 parts)					
BAT-AUX-FRM-3	Expansion Frame 3 — Welded steel frame for rack 3	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-3	Coolant Plate 3 — Liquid cooling plate 3	Buy	—	25.00 kg	£450
BAT-AUX-SENS-3	Voltage Sensor 3 — High precision voltage tap 3	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-30 (3 parts)					
BAT-AUX-FRM-30	Expansion Frame 30 — Welded steel frame for rack 30	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-30	Coolant Plate 30 — Liquid cooling plate 30	Buy	—	25.00 kg	£450
BAT-AUX-SENS-30	Voltage Sensor 30 — High precision voltage tap 30	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-31 (3 parts)					
BAT-AUX-FRM-31	Expansion Frame 31 — Welded steel frame for rack 31	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200

BAT-AUX-COOL-31	Coolant Plate 31 — Liquid cooling plate 31	Buy	—	25.00 kg	£450
BAT-AUX-SENS-31	Voltage Sensor 31 — High precision voltage tap 31	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-32 (3 parts)					
BAT-AUX-FRM-32	Expansion Frame 32 — Welded steel frame for rack 32	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-32	Coolant Plate 32 — Liquid cooling plate 32	Buy	—	25.00 kg	£450
BAT-AUX-SENS-32	Voltage Sensor 32 — High precision voltage tap 32	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-33 (3 parts)					
BAT-AUX-FRM-33	Expansion Frame 33 — Welded steel frame for rack 33	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-33	Coolant Plate 33 — Liquid cooling plate 33	Buy	—	25.00 kg	£450
BAT-AUX-SENS-33	Voltage Sensor 33 — High precision voltage tap 33	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-34 (3 parts)					
BAT-AUX-FRM-34	Expansion Frame 34 — Welded steel frame for rack 34	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-34	Coolant Plate 34 — Liquid cooling plate 34	Buy	—	25.00 kg	£450
BAT-AUX-SENS-34	Voltage Sensor 34 — High precision voltage tap 34	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-35 (3 parts)					
BAT-AUX-FRM-35	Expansion Frame 35 — Welded steel frame for rack 35	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-35	Coolant Plate 35 — Liquid cooling plate 35	Buy	—	25.00 kg	£450
BAT-AUX-SENS-35	Voltage Sensor 35 — High precision voltage tap 35	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-36 (3 parts)					
BAT-AUX-FRM-36	Expansion Frame 36 — Welded steel frame for rack 36	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-36	Coolant Plate 36 — Liquid cooling plate 36	Buy	—	25.00 kg	£450
BAT-AUX-SENS-36	Voltage Sensor 36 — High precision voltage tap 36	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-37 (3 parts)					
BAT-AUX-FRM-37	Expansion Frame 37 — Welded steel frame for rack 37	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-37	Coolant Plate 37 — Liquid cooling plate 37	Buy	—	25.00 kg	£450
BAT-AUX-SENS-37	Voltage Sensor 37 — High precision voltage tap 37	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-38 (3 parts)					

BAT-AUX-FRM-38	Expansion Frame 38 — Welded steel frame for rack 38	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-38	Coolant Plate 38 — Liquid cooling plate 38	Buy	—	25.00 kg	£450
BAT-AUX-SENS-38	Voltage Sensor 38 — High precision voltage tap 38	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-39 (3 parts)					
BAT-AUX-FRM-39	Expansion Frame 39 — Welded steel frame for rack 39	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-39	Coolant Plate 39 — Liquid cooling plate 39	Buy	—	25.00 kg	£450
BAT-AUX-SENS-39	Voltage Sensor 39 — High precision voltage tap 39	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-4 (3 parts)					
BAT-AUX-FRM-4	Expansion Frame 4 — Welded steel frame for rack 4	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-4	Coolant Plate 4 — Liquid cooling plate 4	Buy	—	25.00 kg	£450
BAT-AUX-SENS-4	Voltage Sensor 4 — High precision voltage tap 4	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-5 (3 parts)					
BAT-AUX-FRM-5	Expansion Frame 5 — Welded steel frame for rack 5	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-5	Coolant Plate 5 — Liquid cooling plate 5	Buy	—	25.00 kg	£450
BAT-AUX-SENS-5	Voltage Sensor 5 — High precision voltage tap 5	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-6 (3 parts)					
BAT-AUX-FRM-6	Expansion Frame 6 — Welded steel frame for rack 6	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-6	Coolant Plate 6 — Liquid cooling plate 6	Buy	—	25.00 kg	£450
BAT-AUX-SENS-6	Voltage Sensor 6 — High precision voltage tap 6	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-7 (3 parts)					
BAT-AUX-FRM-7	Expansion Frame 7 — Welded steel frame for rack 7	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-7	Coolant Plate 7 — Liquid cooling plate 7	Buy	—	25.00 kg	£450
BAT-AUX-SENS-7	Voltage Sensor 7 — High precision voltage tap 7	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-8 (3 parts)					
BAT-AUX-FRM-8	Expansion Frame 8 — Welded steel frame for rack 8	Make	Laser cutting & Welding · S275 Steel	150.00 kg	£1,200
BAT-AUX-COOL-8	Coolant Plate 8 — Liquid cooling plate 8	Buy	—	25.00 kg	£450
BAT-AUX-SENS-8	Voltage Sensor 8 — High precision voltage tap 8	Buy	—	0.20 kg	£85
Auxiliary Rack System Alpha-9 (3 parts)					
BAT-AUX-FRM-9		Make		150.00 kg	£1,200

	Expansion Frame 9 — Welded steel frame for rack 9		Laser cutting & Welding · S275 Steel		
BAT-AUX-COOL-9	Coolant Plate 9 — Liquid cooling plate 9	Buy	—	25.00 kg	£450
BAT-AUX-SENS-9	Voltage Sensor 9 — High precision voltage tap 9	Buy	—	0.20 kg	£85
Bidirectional PCS Inverter (2 parts)					
INV-SIC-1MW	SMA 500kW PCS Unit — Grid-forming bidirectional inverter.	Buy	—	1100.00 kg	£85,000
INV-LCL-FLT	LCL Output Filter — Harmonic suppression filter.	Buy	—	85.00 kg	£3,200
BMS / SCADA Control Panel (2 parts)					
BMS-MST-CTRL	Master BMS Controller — System-level battery management unit.	Buy	—	2.50 kg	£3,500
BMS-HMI-15	15" Touchscreen HMI — Local operator interface.	Buy	—	4.20 kg	£1,200
Container Enclosure & Structure (2 parts)					
ENC-40FT-HC	40ft ISO High-Cube Steel Container — Modified ISO 1AAA container with reinforced floor.	Buy	—	3800.00 kg	£12,000
ENC-INS-RW	Rockwool Fire Insulation — 50mm thermal and fire-rated lining.	Buy	—	450.00 kg	£2,500
DC Combiner & Distribution Panel (2 parts)					
DCC-BUS-80	Tinned Copper Busbar 80x10mm — Main aggregation busbar.	Make	Machining · Copper	45.00 kg	£450
DCC-FUSE-1500V	NH2 gPV Fuse 1500V — String isolation fuse.	Buy	—	2.10 kg	£85
Fire Detection & Clean-Agent Suppression (2 parts)					
FIRE-NOVEC-CYL	Novec 1230 Cylinder — Clean-agent suppression tank.	Buy	—	150.00 kg	£5,400
FIRE-SENS-H2	H2 Off-gas Sensor — Early warning off-gas detector.	Buy	—	0.50 kg	£800
HVAC Thermal Management System (2 parts)					
HVAC-40KW-EXT	45kW Packaged Chiller — Rooftop DX chiller unit.	Buy	—	280.00 kg	£18,000
HVAC-DCT-GALV	Galvanised Ducting — Internal air distribution.	Make	Bending · Galvanised Steel	40.00 kg	£600
Port-side LFP Battery Rack Array (2 parts)					
BAT-LFP-280	CATL 280Ah LFP Prismatic Cell — High cycle life LFP cell.	Buy	—	5.50 kg	£120
BAT-RACK-FRM	Rack Steel Frame — Welded and powder-coated steel frame.	Make	Laser & Brake · S275 Steel	150.00 kg	£800
Starboard-side LFP Battery Rack Array (2 parts)					
BAT-LFP-280-SB	CATL 280Ah LFP Prismatic Cell — High cycle life LFP cell.	Buy	—	5.50 kg	£120
BAT-RACK-FRM-SB	Rack Steel Frame — Welded and powder-coated steel frame.	Make	Laser & Brake · S275 Steel	150.00 kg	£800

Total cost	**£227,475**
Total mass	15690.30 kg
Buy / make	94 buy · 44 make

ÁDB

5. Cost waterfall

Cost estimates derived from the module decomposition and the project's cost-estimate data. Per-module figures roll up to the unit cost shown on the cover page.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£494,029	**£180,000**	**£314,029OVER**

Per-module roll-up

MODULE	COST	% OF UNIT
Container Enclosure & Structure	£41,740	8.4%
Lithium Iron Phosphate Battery Rack Assembly	£26,703	5.4%
Bi-directional Grid-Forming Power Conversion System	£287,300	58.2%
Liquid Cooling Skid	£87,370	17.7%
Battery Management System	£10,660	2.2%
Fire Detection and Suppression System	£23,235	4.7%
Auxiliary Power and Low-Voltage Distribution	£17,021	3.4%
Controls and Supervisory Cabinet	£13,300	2.7%